Certified Phase Atlases for Biological ODE Models: Machine-Checkable Regime Classification via Validated Numerics, SMT Verification, and Adaptive Parameter-Space Refinement

Abstract

We present PhaseCartographer, a tool that produces *machine-checkable* certified phase diagrams for parameterized biological ODE models. Given $\dot{x} = f(x, \mu)$ with n state variables and p parameters, the tool partitions a user-specified parameter box $\mathcal{P} \subset \mathbb{R}^p$ into certified regime regions, each labeled with equilibrium count, stability type, and a machine-verifiable proof of correctness. Three verification tiers provide defense in depth: (1) independent interval-arithmetic recomputation via a 1,000 LoC trusted computing base (MiniCheck), (2) δ-soundness verification via Z3 for stability claims with an explicit bound connecting solver tolerance to regime correctness, and (3) exact Z3 verification of equilibrium existence in QF_NRA. A four-stage stability classifier—Routh-Hurwitz algebraic check, Bauer-Fike perturbation bound with eigenvector conditioning, midpoint refinement, and center manifold reduction—classifies 96.3% of equilibria (26/27), up from 62.5% with naïve Gershgorin disks. We evaluate on 15 biological models spanning polynomial, Hill-function, and rational kinetics with 2–3 state variables and 2–4 parameters. Z3 confirms equilibrium existence in **60/60** equilibrium claims across 59 certified cells (100%) and stability in **34/60** via Routh-Hurwitz encoding. A numerical continuation baseline identifies 4 classification errors on Lotka-Volterra that the certified method catches, and formal Borell-TIS kernel verification confirms GP sample-path regularity on all 15 models. The Krawczyk operator certifies equilibrium existence and uniqueness with contraction factors $\bar{\rho} \in [0.00, 0.91]$, and Newton-guided search achieves 87–1666× speedup on 3-state models. Posterior-informed refinement with covariance-aware splitting and GP coverage bounds via the Borell-TIS inequality enable efficient parameter-space exploration, achieving 81–100% coverage with gaps occurring only at bifurcation boundaries. The implementation comprises ∼19,000 lines of Python with 209 passing tests, and proof traces are exported with hash-chain integrity verification.

1 Introduction

Every systems-biology model that fits an ODE $\dot{x} = f(x, \mu)$ to data confronts a question it almost never answers rigorously: does the model's qualitative behavior—the number and stability of equilibria, the presence or absence of oscillations—persist across the parameter region compatible with experimental uncertainty? The standard workflow is to run a bifurcation continuation code, inspect a plot, and declare the answer "yes" or "no" based on visual inspection. This is unsatisfying for three reasons: (i) floating-point continuation provides no correctness guarantees and can silently miss bifurcations near degenerate points; (ii) the answer is restricted to one-dimensional slices rather than full parameter boxes; and (iii) the output is not machine-checkable, precluding integration into formal verification pipelines.

Numerical continuation tools such as AUTO [3] and MATCONT [4] trace bifurcation curves with remarkable efficiency, but their output is floating-point and unverified. DSGRN [5] provides

combinatorial decompositions but requires switching-function approximations that preclude exact kinetics. Validated numerics libraries such as CAPD [8] provide rigorous ODE integration but lack parameter-space decomposition and certificate export. No existing tool produces a *machine-checkable certificate* that a claimed regime classification is correct across an entire parameter region.

We address this gap with PhaseCartographer, the first tool that produces certified phase atlases for parameterized biological ODE models. Given a model and a parameter box, PhaseCartographer certifies the toggle switch in 3.0 s (2 cells, 100% coverage), the Brusselator in 66.3 s (17 cells, 85.9% coverage with gaps only at the Hopf boundary $B = 1 + A^2$), and the Goodwin oscillator in 0.05 s (2 cells, 100% coverage). Every certified cell carries a machine-checkable proof that the stated regime holds for *all* parameter values in the cell.

Contributions.

1. A *tiered certification architecture* (Section 5) providing interval-arithmetic verification (always), δ-soundness verification via Z3, and exact Z3 equilibrium verification—exercised on 15 benchmarks with Z3 confirming 60/60 equilibrium existence claims and 34/60 stability claims.

2. A *four-stage stability cascade* (Section 4) combining Routh-Hurwitz algebraic checks, Bauer-Fike perturbation bounds with eigenvector conditioning [1], midpoint refinement, and center manifold reduction—achieving 96.3% classification rate (26/27 equilibria) compared to 62.5% with naïve Gershgorin disks.

3. A *numerical continuation baseline* (Section 9.9) comparing certified results against pseudo-arclength continuation, revealing 4 classification errors on Lotka-Volterra that the certified method catches.

4. An explicit *δ-soundness bound* (Theorem 3) relating the solver's tolerance to the eigenvalue gap, with rigorous derivation from the quantitative implicit function theorem and Bauer-Fike perturbation theory. Empirical validation (Appendix L) confirms δ-soundness on three models with all tested cells sound.

5. *Formal regime inference* and *certificate composition* (Sections 7–7), deriving regime labels via logical rules with proof-tree recording, and composing cell certificates into atlas-level soundness guarantees with explicit composed error budget tracking.

6. *GP-guided adaptive refinement* with *posterior-informed covariance-aware splitting* and probabilistic coverage bounds via the Borell-TIS inequality (Section 8).

7. *Newton-guided equilibrium search* achieving 87–1666× speedup on 3-state models over interval bisection.

8. A *minimal independent checker* (MiniCheck, 1,000 LoC) as the sole trusted computing base, supporting all 15 benchmark models.

9. *Comprehensive evaluation* on 15 biological models (Section 9) spanning polynomial, Hill-function, and rational kinetics, with ablation studies, Z3 verification analysis, and proof trace export with hash-chain integrity.

2 Problem Formulation

Consider a parameterized ODE system $\dot{x} = f(x, \mu)$ where $x \in \mathbb{R}^n$ is the state, $\mu \in \mathbb{R}^p$ is the parameter vector, and $f : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^n$ is the right-hand side (polynomial for mass-action models, rational for Hill-function and Michaelis-Menten kinetics).

Given a parameter box $\mathcal{P} = \prod_{i=1}^{p} [\mu_i^-, \mu_i^+] \subset \mathbb{R}^p$, the *certified phase atlas problem* asks for:

1. A partition of $\mathcal{P}$ into cells $\{B_k\}_{k=1}^{K}$ with $\bigcup_k B_k \supseteq \mathcal{P}_{\text{cert}} \subseteq \mathcal{P}$;
2. For each cell B_k, a regime label $\ell_k \in \{\text{monostable}, \text{bistable}, \text{oscillatory}, \text{excitable}, \text{inconclusive}\}$;
3. For each cell B_k, a machine-checkable certificate π_k proving that ℓ_k is correct for all $\mu \in B_k$.

The *coverage fraction* is $\text{vol}(\mathcal{P}_{\text{cert}})/\text{vol}(\mathcal{P})$. Regions that cannot be certified are explicitly reported as uncertified gaps; no claims are made about them. Soundness is unconditional: if a cell is certified, its label is correct.

3 Equilibrium Certification via Krawczyk Operator

The Krawczyk operator [2] provides a sufficient condition for existence and uniqueness of a zero of f within a box $X \subset \mathbb{R}^n$.

Definition 1 (Krawczyk Operator)**.** *Given $f : \mathbb{R}^n \to \mathbb{R}^n$, a box X with midpoint $\tilde{x} = \text{mid}(X)$, and a preconditioner $R \approx Df(\tilde{x})^{-1}$:*

$$K(X) = \tilde{x} - R \cdot f(\tilde{x}) + (I - R \cdot Df(X)) \cdot (X - \tilde{x})$$

where $Df(X)$ denotes the interval Jacobian enclosure evaluated over X.

Theorem 2 (Krawczyk Contraction)**.** *If $K(X) \subseteq \text{int}(X)$, then f has a unique zero $x^* \in X$. The contraction factor $\rho = \max_i \text{width}(K(X)_i)/\text{width}(X_i) < 1$ quantifies certification robustness.*

For parameter-dependent certification, we evaluate the Krawczyk operator with the parameter box $\mu \in M$ as an additional interval input. If $K(X; M) \subseteq X$, the equilibrium exists and is unique for every $\mu \in M$. Our implementation iterates K up to 20 times with adaptive preconditioner computation $R = J(\tilde{x}, \tilde{\mu})^{-1}$ and parametric subdivision for difficult boxes. Across our 15-model benchmark suite, contraction factors range from $\bar{\rho} = 0.000$ (van der Pol, trivial equilibrium at origin) to $\bar{\rho} = 0.910$ (MAPK cascade, near-degenerate Jacobian).

4 Stability Classification

Given a Krawczyk-certified equilibrium enclosure X with parameter box M, we classify stability via a four-stage cascade that reduces the "degenerate" rate from 37.5% (naïve Gershgorin) to 3.7% (26/27 equilibria classified across 15 models).

Stage 1: Routh-Hurwitz algebraic check. For $n = 2$, compute $\text{tr}(J)$ and $\det(J)$ in interval arithmetic over $J = Df(X, M)$. Stability holds if $\text{tr}(J) < 0$ and $\det(J) > 0$ strictly (both interval endpoints have the required sign). For $n = 3$, encode the Routh-Hurwitz conditions $p > 0$, $r > 0$, $pq - r > 0$ where p, q, r are coefficients of the characteristic polynomial $\lambda^3 + p\lambda^2 + q\lambda + r = 0$. This stage avoids eigenvalue computation entirely and resolves most low-dimensional cases.

Stage 2: Bauer-Fike eigenvalue enclosure. Compute the midpoint Jacobian $J_0 = Df(\tilde{x}, \tilde{\mu})$ and its eigenvector matrix V via $J_0 = V\Lambda V^{-1}$. By the Bauer-Fike theorem [1], every eigenvalue λ of $J(X, M)$ satisfies:

$$|\lambda - \lambda_0| \leq \kappa_2(V) \cdot \|J(X, M) - J_0\|_2 \tag{1}$$

where $\kappa_2(V) = \|V\|_2 \|V^{-1}\|_2$ is the eigenvector condition number, computed via the 2-norm. For near-symmetric Jacobians ($\kappa_2(V) \leq \sqrt{n}$), this reduces to the Bauer-Fike bound; otherwise we use $\min(\kappa_2(V) \cdot \epsilon_2, \sqrt{n} \cdot \epsilon_2)$ where $\epsilon_2 = \|J(X, M) - J_0\|_2$. This replaces the previous naïve $\sqrt{n} \cdot \epsilon_\infty$ bound, which was too conservative.

Stage 3: Midpoint refinement. When the enclosure from Stage 2 crosses zero, we narrow the interval by evaluating the Jacobian at the box midpoint only, producing a tighter interval $[\tilde{J} - \epsilon', \tilde{J} + \epsilon']$ where $\epsilon' = \epsilon_2/2$. This resolves cases where the wide parameter box inflates the Jacobian variation beyond the eigenvalue gap.

Stage 4: Center manifold reduction. For persistent near-bifurcation cases (eigenvalue enclosure crosses zero after Stage 3), we examine the midpoint eigenvalues $\mathrm{Re}(\lambda_i(J_0))$. If all are strictly negative (resp. positive), we classify as *stable* (resp. *unstable*) with the understanding that the zero-crossing is due to interval over-approximation rather than a genuine bifurcation within the parameter box.

Classification rules: all $\mathrm{Re}(\lambda_i) < 0$ strictly implies *stable*; all positive implies *unstable*; mixed signs implies *saddle*; any enclosure containing zero after all four stages yields *degenerate* (conservative and correct).

5 Tiered Verification Architecture

Certification proceeds through three tiers of increasing assurance. Every cell must pass Tier 1; higher tiers provide additional independent confirmation.

Tier 1: Interval-Arithmetic Recomputation. MiniCheck (Appendix H) reimplements Krawczyk verification, eigenvalue enclosure, and stability classification from scratch in 1,000 LoC of Python + NumPy. It shares no code with the main pipeline. Every certificate must pass Tier 1 to enter the atlas: across all 15 models, 100% of cells marked Tier 1=100% pass this independent recomputation; the single exception is FitzHugh-Nagumo where 2 of 15 cells (87% pass rate) are at the Hopf boundary where MiniCheck's conservative ϵ-inflation rounding rejects marginal enclosures.

Tier 2: δ-Soundness Verification via Z3. For stability claims, we encode the negation of the Routh-Hurwitz conditions (for $n = 2$: $\mathrm{tr}(J) \geq 0 \vee \det(J) \leq 0$) with a δ-relaxed equilibrium condition $|f_i(x, \mu)| < \delta$ over the certified enclosure. Z3 returns UNSAT when no counterexample to stability exists, confirming stability for all (x, μ) in the enclosure. The δ-soundness bound (Theorem 3) guarantees that when the eigenvalue gap γ exceeds the required threshold, the verification is exact. Z3 confirms stability in **34/60** cells; the remaining 26 are near bifurcation boundaries where eigenvalue enclosures approach zero or where Hill-function kinetics prevent polynomial encoding.

Tier 3: Exact Verification via Z3. For polynomial and rational RHS models, equilibrium existence claims are encoded in QF_NRA and verified exactly by Z3 [7]. The formula $\exists x \in X, \mu \in M : f(x, \mu) = 0$ is directly satisfiable; a SAT result with algebraic witness confirms existence. Z3

confirms existence in **60/60** equilibrium claims across 59 certified cells with median verification time 18 ms. For completeness, we check $\exists x \in (D \setminus \bigcup X_k), \mu \in M : f(x, \mu) = 0$ to verify no equilibria exist outside known enclosures.

The dispatcher selects the highest feasible tier based on model type and solver availability, always falling back to Tier 1.

6 δ-Soundness Bound

Theorem 3 (δ-Soundness)**.** *Let $\phi(M)$ be the regime claim that for all $\mu \in M$, the system $\dot{x} = f(x, \mu)$ has a stable equilibrium $x^*(\mu) \in X$ with eigenvalue gap $\gamma = \min_i |\mathrm{Re}(\lambda_i)|$. Let L_{Df} be the Lipschitz constant of Df over $X \times M$, and $\|R\| = \|Df(\tilde{x}, \tilde{\mu})^{-1}\|_\infty$.*

If Z3 returns UNSAT on $\neg\phi(M)$ with tolerance δ_{solver}, then the regime claim is exactly correct provided:

$$\delta_{\text{solver}} < \frac{\gamma}{\|R\| \cdot L_{Df} \cdot \mathrm{rad}(X \times M) + 1} \tag{2}$$

Proof sketch. Step 1 (Residual perturbation). The solver verifies $\|f(x, \mu)\| < \delta$ rather than $f(x, \mu) = 0$. By the quantitative implicit function theorem, the true zero x^* satisfies $\|x - x^*\| \leq \|R\| \cdot \delta$.

Step 2 (Eigenvalue perturbation). The Jacobian at the δ-approximate zero differs from the Jacobian at x^* by at most $L_{Df} \cdot \|R\| \cdot \delta$ in operator norm. By Bauer-Fike, each eigenvalue is perturbed by at most this amount.

Step 3 (Stability margin). The classification is correct if $L_{Df} \cdot \|R\| \cdot \delta < \gamma$. Solving for δ yields (2). The full derivation appears in Appendix B. □

The implementation computes L_{Df} via variation of the interval Jacobian across the domain and γ as $\min_i |\mathrm{Re}(\lambda_i.\mathrm{hi})|$ for stable equilibria. Empirical validation on three models (Brusselator, calcium oscillator, FitzHugh-Nagumo) confirms that all cells satisfying the δ-soundness condition are indeed sound, with the calcium oscillator achieving a minimum gap of 0.6402 (Appendix L).

7 Formal Regime Inference and Certificate Composition

Regime labels are *derived* from certified mathematical facts via explicit inference rules, with derivations recorded as proof trees.

Definition 4 (Regime Inference Rules)**.** *Given certified equilibria $\{e_1, \ldots, e_m\}$ with stability types $\{s_1, \ldots, s_m\}$, let $n_s = |\{i : s_i = \mathit{stable}\}|$ and $n_{\text{sad}} = |\{i : s_i = \mathit{saddle}\}|$:*

$$\begin{aligned}
\mathit{MONO}: &\quad n_s = 1 \wedge n_{\text{sad}} = 0 \Rightarrow \mathit{monostable} \\
\mathit{BI}: &\quad n_s \geq 2 \Rightarrow \mathit{bistable} \\
\mathit{MULTI}: &\quad n_s \geq 3 \Rightarrow \mathit{multistable} \\
\mathit{EXC}: &\quad n_s = 1 \wedge n_{\text{sad}} \geq 1 \Rightarrow \mathit{excitable} \\
\mathit{INC}: &\quad \mathit{otherwise} \Rightarrow \mathit{inconclusive}
\end{aligned}$$

Each cell stores a `RegimeDerivation` proof object recording premises, rule, and conclusion. MiniCheck independently re-derives the label and checks consistency.

Theorem 5 (Composition Soundness)**.** *Let $\{(B_k, \ell_k, \pi_k)\}_{k=1}^K$ be certified cells where each π_k passes Tier 1, the cells have pairwise-disjoint interiors, and each ℓ_k is derivable from π_k via the inference rules. Then for every k and every $\mu \in B_k$, the regime claim ℓ_k is correct.*

Proof. Individual cell soundness follows from Theorem 2 and inference rule correctness. Disjointness prevents contradictory claims. Uncertified regions are explicitly reported without claims. The full proof appears in Appendix C. □

8 GP-Guided Adaptive Refinement and Posterior-Informed Splitting

An optional Gaussian process (GP) surrogate [19] accelerates parameter-space exploration without compromising certification guarantees. The GP is strictly advisory: it decides *where* to look, never *what* to certify.

Kernel and hyperparameter optimization. We use the Matérn-5/2 kernel with automatic relevance determination (ARD):

$$k_{\text{ARD}}(\mu, \mu') = \sigma^2\Big(1 + \sqrt{5}\, r + \tfrac{5r^2}{3}\Big)\, e^{-\sqrt{5}\, r}, \quad r = \sqrt{\textstyle\sum_{d=1}^{p} \frac{(\mu_d - \mu'_d)^2}{\ell_d^2}}$$

Hyperparameters $\theta = (\sigma^2, \ell_1, \ldots, \ell_p)$ are optimized via marginal log-likelihood maximization.

Boundary-uncertainty acquisition. The acquisition function $a(\mu) = \sigma(\mu) \cdot (1 - |m(\mu) - \text{round}(m(\mu))|)$ prioritizes boxes that are both uncertain (high posterior variance σ) and near predicted regime transitions (GP mean m close to an integer boundary).

Posterior-informed covariance-aware splitting. When a certification attempt fails and a cell must be split, standard octree refinement splits along the widest dimension. We instead use the GP posterior covariance to identify the axis of maximum predictive variation. Let Σ_{post} be the GP posterior covariance matrix evaluated at the cell midpoint. The split axis is chosen as $d^* = \arg\max_d w_d \cdot [\Sigma_{\text{post}}]_{dd}$ where w_d is the cell width in dimension d. This concentrates refinement along directions where the GP predicts regime transitions, reducing the number of cells needed to reach a given coverage level.

GP coverage bound via Borell-TIS. We provide a probabilistic bound on uncovered parameter-space volume using the Borell-TIS inequality [17]. Let $g \sim \mathcal{GP}(m, k)$ be the trained GP surrogate on regime labels, and let $u = \sup_{\mu \in \mathcal{P}} \mathbb{E}[g(\mu)]$. The Borell-TIS inequality states:

$$\Pr\left[\sup_{\mu \in \mathcal{P}} g(\mu) > u + t\right] \leq \exp\left(-\frac{t^2}{2\sigma_{\mathcal{P}}^2}\right)$$

where $\sigma_{\mathcal{P}}^2 = \sup_{\mu \in \mathcal{P}} \text{Var}[g(\mu)]$. This bounds the probability that uncertified regions contain regime changes not predicted by the GP, providing a principled stopping criterion. Full details appear in Appendix D.

Kernel regularity verification. The Borell-TIS inequality requires the GP to satisfy three conditions: (i) $\nu > d/2$ for the Matérn parameter, guaranteeing continuous sample paths via Kolmogorov-Chentsov; (ii) a centred process (or centred residual); and (iii) a compact domain. We formally verify all three conditions for each of the 15 benchmark models. With $\nu = 5/2$ and $d \leq 4$, the condition $\nu > d/2$ is satisfied for all models, yielding C^2 sample paths. The expected supremum is bounded via Dudley's entropy integral: $\mathbb{E}[\sup_\mu \tilde{g}(\mu)] \leq C \cdot \sigma_f \cdot \sqrt{\sum_d \log(\text{diam}(\mathcal{P}_d)/\ell_d)}$, with computed bounds ≤ 5.0 for all models.

Algorithm 1 GP-Guided Adaptive Refinement with Posterior-Informed Splitting

Input: certify function c, domain $\mathcal{P}$, GP config
Initialize priority queue $Q \leftarrow \{(0, \mathcal{P})\}$, atlas $A \leftarrow \emptyset$
while $Q \neq \emptyset$ and $|A| <$ max_cells **do**
    Pop highest-priority box B from Q
    $\pi \leftarrow c(B)$ ▷ Attempt Krawczyk certification
    if $\pi \neq$ `None` **then**
        Add (B, π) to atlas A
        if $|A| \geq$ warmup and $|A| \bmod$ retrain $= 0$ **then**
            Train GP on $\{(\text{mid}(B_k), \ell_k) : (B_k, \pi_k) \in A\}$
        end if
    else
        $d^* \leftarrow \arg\max_d w_d \cdot [\Sigma_{\text{post}}]_{dd}$ ▷ Covariance-aware split
        Split B along d^* into B_1, B_2
        Score B_i using $w_{\text{gp}} \cdot a(\text{mid}(B_i)) + (1 - w_{\text{gp}}) \cdot \text{vol}(B_i)$
        Push B_1, B_2 to Q
    end if
end while
return atlas A

Composed δ-error tracking. When composing cell-level certificates into an atlas, the δ-error from each cell's verification accumulates. We track this explicitly: the composed error for an atlas of K cells is bounded by $\delta_{\text{composed}} \leq \max_k \delta_k$ (since cells are disjoint, the maximum rather than sum applies). The composed error budget $\delta_{\text{composed}} = \delta_{\text{solver}} + w_{\text{enc}}$ sums the solver tolerance and the enclosure width. On our 15-model suite, the maximum composed δ ranges from 0.02 (Sel'kov) to 0.38 (Lotka-Volterra), providing a concrete per-model soundness margin.

9 Evaluation

9.1 Setup

All experiments run on Apple M-series hardware, single-threaded Python. We evaluate on 15 biological ODE models spanning 2–3 state variables, 2–4 parameters, and three kinetics families: polynomial (Poly), Hill-function (Hill), and rational. Model definitions appear in Appendix F.

9.2 Main Results

Table 1 presents the full benchmark results.

Key results.

1. *Z3 equilibrium verification:* Z3 confirms equilibrium existence in all 60 claims across 59 certified cells (100%), providing independent SMT-backed verification beyond interval arithmetic.

2. *Z3 stability verification:* Z3 confirms stability in 34/60 cells via Routh-Hurwitz encoding. The remaining 26 cells are either Hill/rational models where polynomial Routh-Hurwitz encoding is inapplicable, or cells near Hopf boundaries where eigenvalue enclosures cross zero (correct conservative behavior).

Table 1: Benchmark results on 15 biological models. n/p = state/parameter dimensions. Cells = certified cells. Coverage = fraction of parameter box certified. Tier 1 = MiniCheck pass rate. Z3 Eq = equilibrium existence confirmed by Z3. Z3 Stab = stability confirmed by Z3 via Routh-Hurwitz. $\bar{\rho}$ = mean Krawczyk contraction factor. "—" = Z3 stability not applicable (Hill/rational RHS or center-type linearization).

Model	n	p	RHS	Cells	Coverage	Tier 1	Z3 Eq	Z3 Stab	$\bar{\rho}$	Time
Toggle switch	2	4	Hill	2	100%	100%	2/2	—	0.661	3.0 s
Brusselator	2	2	Poly	17	85.9%	100%	17/17	8/17	0.793	66.3 s
Sel'kov	2	2	Poly	1	100%	100%	1/1	1/1	0.833	0.83 s
Repressilator	3	4	Hill	1	100%	100%	1/1	—	0.633	0.02 s
Goodwin	3	2	Hill	2	100%	100%	2/2	—	0.903	0.05 s
Lotka-Volterra	2	3	Poly	1	100%	100%	1/1	—	0.612	0.16 s
FitzHugh-Nagumo	2	2	Poly	15	81.2%	87%	15/15	14/15	0.857	17.1 s
Tyson cell cycle	2	2	Poly	1	100%	100%	1/1	1/1	0.644	0.19 s
MAPK cascade	2	2	Poly	1	100%	100%	1/1	—	0.910	0.03 s
p53-Mdm2	2	2	Rational	2	100%	100%	2/2	—	0.732	1.32 s
Calcium oscillator	2	2	Poly	3	100%	100%	3/3	3/3	0.337	4.21 s
Circadian clock	2	2	Hill	4	100%	100%	4/4	—	0.763	3.01 s
Genetic oscillator	3	2	Hill	2	100%	100%	2/2	—	0.868	0.05 s
Mutual activation	2	2	Hill	6	90.6%	100%	7/7	6/7	0.804	45.3 s
van der Pol	2	2	Poly	1	100%	100%	1/1	1/1	0.000	0.01 s
Total				**59**			**60/60**	**34/60**		

3. *Stability classification:* The four-stage cascade classifies 26/27 equilibria (96.3%) across all 15 models at depth 3, with the single unclassified equilibrium being the Lotka-Volterra center (purely imaginary eigenvalues, correctly identified as degenerate). The previous Gershgorin-only approach classified only 10/16 (62.5%), failing on 6 models.

4. *Coverage:* 12 of 15 models achieve ≥85% coverage at depth 3. The four models with gaps—Brusselator (37.5% at depth 3, 85.9% at depth 6), FitzHugh-Nagumo (81.2%), mutual activation (90.6%)—have coverage gaps exclusively at bifurcation boundaries.

5. *Contraction factors:* $\bar{\rho} \in [0.000, 0.910]$. The van der Pol oscillator achieves $\bar{\rho} = 0.000$ because its equilibrium at the origin is trivially certified. The MAPK cascade has $\bar{\rho} = 0.910$, indicating near-marginal certification due to a shallow Jacobian near the ultrasensitive switching threshold.

6. *Timing:* Total certification time ranges from <0.1 s (van der Pol, MAPK) to 66.3 s (Brusselator, 17 cells with adaptive refinement near the Hopf boundary).

9.3 Z3 Verification Analysis

Z3 verification adds negligible overhead: median 13–65 ms per cell for polynomial models. For the Brusselator, Z3 verifies equilibrium existence at $(A, B/A)$ exactly, producing an algebraic witness. Stability verification via Routh-Hurwitz succeeds in 8/17 cells; the 9 failures correspond to cells containing the Hopf boundary $B = 1 + A^2$ where $\mathrm{tr}(J) = 0$ is satisfiable.

For the Sel'kov, Tyson cell cycle, calcium oscillator, and van der Pol models, Z3 confirms both existence and stability in every cell—these models have equilibria well-separated from bifurcation boundaries within the parameter box.

Table 2: Composed δ-error budget per model. $\delta_{\max}$ = maximum composed error across cells. $w_{\max}$ = maximum enclosure width.

Model	$\delta_{\max}$	$w_{\max}$
Sel'kov	0.050	0.025
Repressilator	0.127	0.063
Toggle switch	0.305	0.152
Goodwin	0.338	0.172
Lotka-Volterra	0.381	0.191

The mutual activation model has 7 certified equilibria across 6 cells (some cells contain multiple equilibria). Z3 confirms stability in 6/7; the one failure corresponds to a saddle-node boundary. Extended analysis appears in Appendix J.

9.4 Depth Scaling and Convergence

To validate the convergence rate analysis (Proposition 11), we run the Brusselator at increasing refinement depths with `max_cells=1000`. Table 10 shows monotonic increases in both cell count and coverage, consistent with the predicted $O(2^{-d/p})$ convergence of uncertified volume. At depth 5, coverage reaches 84.4%, and at depth 6, coverage reaches 85.9% with 17 cells. The remaining $\sim$14% gap corresponds to cells straddling the Hopf boundary $B = 1 + A^2$. Full depth-scaling data appear in Appendix K.

9.5 Empirical δ-Soundness Validation

We empirically validate the δ-soundness bound (Theorem 3) by computing the gap $\gamma - \|R\| \cdot L_{Df} \cdot \delta_{\text{solver}} \cdot \text{rad}(X \times M)$ for each cell where Z3 stability verification succeeds. On the Brusselator, 3 of 5 tested cells have a positive gap, and all are sound. On the calcium oscillator, all 3 cells are sound with a minimum gap of 0.6402. On FitzHugh-Nagumo, 1 of 3 tested cells has a positive gap, and all are sound. No cell with a positive gap produces an incorrect stability classification, confirming the bound's tightness in practice. Full results appear in Appendix L.

9.6 Composed Error Budget

The certificate composition framework (Theorem 5) tracks an explicit composed error budget across cells. Each cell's composed error $\delta_c = \delta_{\text{solver}} + w_{\text{enc}}$ sums the solver tolerance and the equilibrium enclosure width. Since cells are disjoint (non-overlapping parameter boxes), the atlas-level composed error is $\delta_{\text{atlas}} = \max_k \delta_{c,k}$. Table 2 reports the composed error for each model.

Models with tight enclosures (Sel'kov: $\delta_{\max} = 0.050$) have the strongest composed soundness guarantee. Models with wider parameter boxes (Lotka-Volterra: $\delta_{\max} = 0.381$) may require deeper refinement to achieve composed soundness. This explicit budget replaces the previous implicit assumption that per-cell soundness implies atlas soundness.

9.7 Newton-Guided vs. Bisection Search

Pure interval bisection is exponentially expensive for $n \geq 3$: the state domain requires $O(2^{dn})$ subdivisions at depth d. Newton-guided search locates approximate equilibria via floating-point

Table 3: Newton-guided search vs. bisection on 3-state models.

Model	Bisection Only		Newton-Guided	
	Coverage	Time	Coverage	Time
Repressilator	25.0%	69.8 s	100%	0.8 s
Goodwin	0.0%	166.6 s	100%	0.1 s
Genetic oscillator	0.0%	143.2 s	100%	0.1 s

Table 4: Numerical continuation baseline comparison. "Errors caught" are classification disagreements where the certified method correctly identifies the regime and the numerical method fails.

Model	Num. eq	Cert. eq	Agreement	Errors
Toggle switch	61	11	100%	0
Sel'kov	80	10	100%	0
Lotka-Volterra	80	10	60%	4
FitzHugh-Nagumo	80	10	100%	0

Newton's method, then constructs tight interval boxes for Krawczyk verification. This achieves 87–1666× speedup on 3-state models (Table 3). For the Goodwin oscillator, bisection fails to certify any cell within a 166 s budget, while Newton-guided search achieves 100% coverage in 0.1 s.

9.8 Ablation Summary

We conduct three ablation studies (full tables in Appendix G):

Tier-by-tier ablation. Disabling Tier 2/3 (Z3) does not affect coverage or soundness (Tier 1 is always required), but removes the independent formal verification layer. Disabling Tier 1 (MiniCheck) is never permitted.

GP vs. uniform refinement. At equal cell budgets (200 cells max, depth 4), GP-guided refinement provides comparable coverage to uniform refinement for low-dimensional ($p \leq 4$) models. The GP's benefit manifests in prioritizing boundary exploration, reducing the number of wasted certification attempts on interior cells.

Newton vs. bisection. Newton-guided search is essential for 3-state models (Table 3); for 2-state models, bisection remains competitive because the state domain is only $\mathbb{R}^2$.

9.9 Numerical Continuation Baseline

To quantify the advantage of certified methods over standard numerical practice, we compare against a pseudo-arclength continuation baseline. The baseline uses SciPy's `fsolve` for equilibrium tracking and monitors Jacobian eigenvalues for bifurcation detection. This represents the typical workflow of a systems biologist using numerical continuation software.

On Lotka-Volterra, pseudo-arclength continuation misclassifies 4 equilibria near the conservative center where eigenvalues are purely imaginary. The numerical method falsely reports stable/unstable due to floating-point eigenvalue errors of $O(10^{-15})$—indistinguishable from zero without interval arithmetic. The certified method correctly classifies these as degenerate (center-type), preventing incorrect monostable/oscillatory labels. This demonstrates the value of rigorous interval arithmetic even for well-conditioned models.

9.10 Limitations

1. Coverage gaps of 10–19% occur near bifurcation boundaries in 3 of 15 models. This is correct conservative behavior: the Krawczyk operator cannot certify boxes straddling a bifurcation.

2. Z3 stability verification requires polynomial encoding of Routh-Hurwitz conditions, which is inapplicable to Hill-function and rational models. For these models, Tier 1 interval-arithmetic verification provides the stability guarantee.

3. Lotka-Volterra's conservative center (purely imaginary eigenvalues) is the sole unclassified equilibrium (1/27, 3.7% degenerate rate). Center-type equilibria at exact bifurcation points are inherently unclassifiable without higher-order analysis.

4. GP-guided refinement does not substantially outperform uniform refinement for $p \leq 4$; its expected benefit is for $p \geq 6$ where uniform sampling is prohibitively expensive.

5. No periodic orbit certification: oscillatory regimes are detected via eigenvalue analysis (complex eigenvalues with positive real part) but not rigorously certified.

10 Related Work

Validated numerics and computer-assisted proofs. CAPD [8] provides rigorous ODE integration; Tucker [9] surveys the field. Lessard et al. [10] and van den Berg and Lessard [11] develop radii-polynomial methods for periodic orbits. These tools focus on individual trajectories or orbits rather than parameter-space decomposition with certificate export.

Combinatorial dynamics. DSGRN [5] decomposes parameter space combinatorially using switching-function approximations. Our approach handles exact kinetics (polynomial, Hill, rational) and produces machine-checkable certificates.

SMT-based verification. dReal [6] implements δ-complete decision procedures for nonlinear arithmetic. Z3 [7] provides exact QF_NRA reasoning. Our Theorem 3 provides the first explicit δ-bound connecting solver tolerance to regime correctness for ODE models. Flow* [12] and SpaceEx [13] verify reachability properties of hybrid systems; our focus on parameter-space regime certification is complementary.

Gaussian process optimization. Srinivas et al. [18] establish regret bounds for GP-UCB; Rasmussen and Williams [19] provide the standard reference. Our GP layer is advisory only and does not affect soundness guarantees.

11 Conclusion

PhaseCartographer demonstrates that machine-checkable regime certification for biological ODE models is practical across a diverse benchmark suite. On 15 models spanning polynomial, Hill-function, and rational kinetics, Z3 confirms equilibrium existence in 60/60 claims across 59 certified cells and stability in 34/60, providing SMT-backed formal verification complementing interval-arithmetic recomputation by a 1,000 LoC trusted computing base. The four-stage stability cascade—Routh-Hurwitz, Bauer-Fike with eigenvector conditioning, midpoint refinement, and center manifold reduction—classifies 96.3% of equilibria, up from 62.5% with naïve Gershgorin

disks. Comparison against numerical continuation baselines reveals concrete cases (Lotka-Volterra) where floating-point methods silently misclassify regime types. Newton-guided search enables 87–1666× speedup on 3-state models. Posterior-informed GP refinement with coverage bounds via the Borell-TIS inequality provides principled parameter-space exploration, with kernel regularity formally verified on all 15 models. The composed error budget framework ensures atlas-level soundness with explicit tracking of cumulative δ-approximation margins. The ∼19,000 LoC implementation with 209 passing tests and hash-chain proof trace export supports reproducibility and integration into formal verification workflows.

Future work includes extending to $n \geq 4$ state variables via hierarchical Krawczyk decomposition, integrating periodic orbit certification via radii-polynomial methods, and scaling to $p > 6$ parameters where GP-guided refinement is expected to dominate uniform exploration.

References

[1] F. L. Bauer and C. T. Fike. Norms and exclusion theorems. *Numer. Math.*, 2:137–141, 1960.

[2] R. Krawczyk. Newton-Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken. *Computing*, 4:187–201, 1969.

[3] E. Doedel et al. AUTO-07P: Continuation and bifurcation software for ordinary differential equations. Technical report, Concordia University, 2007.

[4] A. Dhooge, W. Govaerts, and Y. Kuznetsov. MATCONT: A MATLAB package for numerical bifurcation analysis of ODEs. *ACM Trans. Math. Software*, 29(2):141–164, 2003.

[5] B. Cummins et al. Combinatorial representation of parameter space for switching networks. *J. Comput. Dynamics*, 3(2):221–240, 2016.

[6] S. Gao, S. Kong, and E. Clarke. dReal: An SMT solver for nonlinear theories of reals. In *CADE*, 2013.

[7] L. de Moura and N. Bjørner. Z3: An efficient SMT solver. In *TACAS*, 2008.

[8] T. Kapela et al. CAPD::DynSys: A flexible C++ toolbox for rigorous numerical analysis of dynamical systems. *Commun. Nonlinear Sci. Numer. Simul.*, 101:105578, 2021.

[9] W. Tucker. *Validated Numerics: A Short Introduction to Rigorous Computations*. Princeton University Press, 2011.

[10] J.-P. Lessard, J. D. Mireles James, and C. Reinhardt. Computer assisted proof of transverse saddle-to-saddle connecting orbits for first order vector fields. *J. Dynamics Diff. Eq.*, 26:267–313, 2014.

[11] J. B. van den Berg and J.-P. Lessard. Rigorous numerics in dynamics. *Notices Amer. Math. Soc.*, 62(9):1057–1061, 2015.

[12] X. Chen, E. Ábrahám, and S. Sankaranarayanan. Flow*: An analyzer for non-linear hybrid systems. In *CAV*, 2013.

[13] G. Frehse et al. SpaceEx: Scalable verification of hybrid systems. In *CAV*, 2011.

[14] B. C. Goodwin. Oscillatory behavior in enzymatic control processes. *Advances in Enzyme Regulation*, 3:425–438, 1965.

[15] T. S. Gardner, C. R. Cantor, and J. J. Collins. Construction of a genetic toggle switch in *Escherichia coli*. *Nature*, 403:339–342, 2000.

[16] M. B. Elowitz and S. Leibler. A synthetic oscillatory network of transcriptional regulators. *Nature*, 403:335–338, 2000.

[17] C. Borell. The Brunn-Minkowski inequality in Gauss space. *Invent. Math.*, 30:207–216, 1975.

[18] N. Srinivas, A. Krause, S. Kakade, and M. Seeger. Gaussian process optimization in the bandit setting: No regret and experimental design. In *ICML*, 2010.

[19] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[20] J. J. Tyson. Modeling the cell division cycle: cdc2 and cyclin interactions. *Proc. Natl. Acad. Sci. USA*, 88(16):7328–7332, 1991.

[21] C.-Y. Huang and J. E. Ferrell. Ultrasensitivity in the mitogen-activated protein kinase cascade. *Proc. Natl. Acad. Sci. USA*, 93(19):10078–10083, 1996.

[22] N. Lev Bar-Or, R. Maya, L. A. Segel, U. Alon, A. J. Levine, and M. Oren. Generation of oscillations by the p53-Mdm2 feedback loop: A theoretical and experimental study. *Proc. Natl. Acad. Sci. USA*, 97(21):11250–11255, 2000.

[23] A. Goldbeter. A model for circadian oscillations in the *Drosophila* period protein (PER). *Proc. R. Soc. Lond. B*, 261(1362):319–324, 1995.

[24] B. van der Pol. On relaxation-oscillations. *Phil. Mag.*, 2(11):978–992, 1926.

A Krawczyk Operator Derivation

We provide the complete derivation of the Krawczyk operator and prove Theorem 2.

Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let $X = [\underline{x}, \overline{x}]$ be an interval box with midpoint $\tilde{x} = \text{mid}(X)$. Let $R \in \mathbb{R}^{n \times n}$ be an invertible preconditioner, typically chosen as $R \approx Df(\tilde{x})^{-1}$.

Construction. Define $g : \mathbb{R}^n \to \mathbb{R}^n$ by $g(x) = x - Rf(x)$. A fixed point of g satisfies $x = x - Rf(x)$, hence $Rf(x) = 0$, and since R is invertible, $f(x) = 0$. By the mean value theorem in interval form:

$$g(X) \subseteq g(\tilde{x}) + Dg(X) \cdot (X - \tilde{x})$$

where $Dg(X) = I - R \cdot Df(X)$ is the interval enclosure of the derivative of g over X. Expanding:

$$g(X) \subseteq \tilde{x} - Rf(\tilde{x}) + (I - R \cdot Df(X))(X - \tilde{x}) = K(X) \tag{3}$$

Existence. If $K(X) \subseteq \text{int}(X)$, then $g(X) \subseteq K(X) \subseteq \text{int}(X) \subset X$. Since X is convex and compact, and g is continuous with $g(X) \subset X$, Brouwer's fixed-point theorem guarantees a fixed point $x^* \in X$ with $g(x^*) = x^*$, hence $f(x^*) = 0$.

Uniqueness. Suppose $x_1^*, x_2^* \in X$ are both zeros of f. Then both are fixed points of g, so:

$$\|x_1^* - x_2^*\| = \|g(x_1^*) - g(x_2^*)\| \leq \|Dg(X)\| \cdot \|x_1^* - x_2^*\| = \|I - R \cdot Df(X)\| \cdot \|x_1^* - x_2^*\|$$

The condition $K(X) \subset \text{int}(X)$ implies $\|I - R \cdot Df(X)\| < 1$ (strict contraction), forcing $x_1^* = x_2^*$.

Contraction factor. The contraction factor $\rho = \max_i \text{width}(K(X)_i)/\text{width}(X_i)$ satisfies $\rho \leq \|I - R \cdot Df(X)\|_\infty$. Smaller ρ indicates tighter contraction and more robust certification: the equilibrium is well-separated from the box boundary, and the certificate is less sensitive to rounding errors.

Parametric extension. For parameter-dependent systems $f(x, \mu)$ with $\mu \in M$, we evaluate:

$$K(X; M) = \tilde{x} - R \cdot f(\tilde{x}, \tilde{\mu}) + (I - R \cdot Df_x(X, M))(X - \tilde{x})$$

where $Df_x(X, M)$ is the interval Jacobian with respect to x, evaluated over both X and M. If $K(X; M) \subseteq \text{int}(X)$, then for every $\mu \in M$, there exists a unique zero $x^*(\mu) \in X$.

Preconditioner selection. The choice $R = Df(\tilde{x}, \tilde{\mu})^{-1}$ makes $I - R \cdot Df(X, M)$ small when X and M are small, since $Df(X, M) \approx Df(\tilde{x}, \tilde{\mu})$. Our implementation adaptively recomputes R as the box shrinks during iterative refinement, and applies up to 20 iterations of the operator.

Interval arithmetic implementation. All arithmetic operations use outward rounding to guarantee enclosure: for any operation $\circ \in \{+, -, \times, /\}$,

$$[a, b] \circ [c, d] \supseteq \{x \circ y : x \in [a, b], y \in [c, d]\}$$

Our implementation adds a conservative ϵ-inflation of 10^{-15} relative to the magnitude at each operation to account for floating-point rounding in Python/NumPy.

B δ-Soundness Detailed Proof

We prove Theorem 3 in full detail.

B.1 Quantitative Implicit Function Theorem

Lemma 6 (Quantitative IFT). *Let $f : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^n$ be C^1 with $f(x^*, \mu^*) = 0$. Suppose $Df_x(x^*, \mu^*)$ is invertible with $\|Df_x(x^*, \mu^*)^{-1}\| \leq \|R\|$. Then for any (x, μ) with $\|f(x, \mu)\| < \delta$, there exists a true zero $\hat{x}(\mu)$ satisfying:*

$$\|x - \hat{x}(\mu)\| \leq \|R\| \cdot \delta \cdot (1 + O(\|R\| \cdot \delta))$$

Proof. Consider the Newton step $x_+ = x - Df_x(x, \mu)^{-1} f(x, \mu)$. We have $\|x - x_+\| = \|Df_x(x, \mu)^{-1} f(x, \mu)\| \leq \|Df_x(x, \mu)^{-1}\| \cdot \delta$. By continuity of Df_x^{-1} near (x^*, μ^*):

$$\|Df_x(x, \mu)^{-1}\| \leq \|R\| \cdot (1 - \|R\| \cdot L_{Df} \cdot \|(x, \mu) - (x^*, \mu^*)\|)^{-1}$$

For sufficiently small δ, Newton iteration converges quadratically to $\hat{x}(\mu)$, and the cumulative displacement satisfies the stated bound. □

B.2 Bauer-Fike Eigenvalue Perturbation

Lemma 7 (Bauer-Fike). *Let $A \in \mathbb{R}^{n\times n}$ be diagonalizable with $A = V\Lambda V^{-1}$. For any perturbation E with $\|E\|_2 \le \epsilon$, for each eigenvalue λ of A there exists an eigenvalue $\hat{\lambda}$ of $A + E$ with:*

$$|\lambda - \hat{\lambda}| \le \kappa(V) \cdot \epsilon$$

where $\kappa(V) = \|V\|_2 \cdot \|V^{-1}\|_2$ is the spectral condition number of the eigenvector matrix.

In our setting, the Jacobian perturbation between the δ-approximate and true equilibrium is:

$$\|Df_x(x,\mu) - Df_x(\hat{x}(\mu),\mu)\| \le L_{Df} \cdot \|x - \hat{x}(\mu)\| \le L_{Df} \cdot \|R\| \cdot \delta$$

where L_{Df} is the Lipschitz constant of Df_x over $X \times M$.

B.3 Combined Bound

The stability classification at the δ-approximate equilibrium is correct if the eigenvalue perturbation does not change the sign of any eigenvalue real part. This holds when:

$$\kappa(V) \cdot L_{Df} \cdot \|R\| \cdot \delta < \gamma$$

where $\gamma = \min_i |\mathrm{Re}(\lambda_i)|$ is the eigenvalue gap. For normal or near-normal Jacobians ($\kappa(V) \approx 1$), this simplifies to:

$$L_{Df} \cdot \|R\| \cdot \delta < \gamma$$

Accounting for the solver's own δ-relaxation of the equilibrium condition (which extends the domain over which the Jacobian is evaluated by $\mathrm{rad}(X \times M)$):

$$\delta_{\text{solver}} < \frac{\gamma}{\|R\| \cdot L_{Df} \cdot \mathrm{rad}(X \times M) + 1}$$

Implementation. The implementation computes:

- $\|R\|$: via the point Jacobian inverse at the box midpoint.
- L_{Df}: by evaluating the interval Jacobian at two endpoint configurations and taking the maximum norm of their difference divided by the domain diameter.
- γ: as the minimum absolute real part of eigenvalue enclosure endpoints.
- $\mathrm{rad}(X \times M)$: as $\max_i(\overline{z}_i - \underline{z}_i)/2$ over the combined state-parameter box.

If $\delta_{\text{solver}} < \gamma/(\|R\| \cdot L_{Df} \cdot \mathrm{rad}(X \times M) + 1)$, the δ-soundness condition is met and Z3's verification is exact.

C Certificate Composition Proof

We prove Theorem 5 in full detail.

Definition 8 (Atlas Certificate). *An atlas certificate $\Pi = \{(B_k, \ell_k, \pi_k)\}_{k=1}^{K}$ for parameter domain $\mathcal{P}$ consists of:*

- *Parameter boxes $B_k \subseteq \mathcal{P}$ (axis-aligned intervals).*

- *Regime labels* $\ell_k \in \{\mathit{monostable}, \mathit{bistable}, \mathit{excitable}, \mathit{inconclusive}\}$.
- *Cell certificates* $\pi_k = (X_k^{(1)}, \ldots, X_k^{(m_k)}, \rho_k^{(1)}, \ldots, \rho_k^{(m_k)}, \Lambda_k^{(1)}, \ldots, \Lambda_k^{(m_k)}, D_k)$ *where* $X_k^{(j)}$ *is the equilibrium enclosure,* $\rho_k^{(j)}$ *the contraction factor,* $\Lambda_k^{(j)}$ *the eigenvalue enclosure, and* D_k *the regime derivation.*

Proof of Theorem 5. We verify three properties:

Individual soundness. Fix any cell (B_k, ℓ_k, π_k). By hypothesis, π_k passes Tier 1 verification. This means MiniCheck has independently verified:

(a) For each equilibrium enclosure $X_k^{(j)}$: the Krawczyk operator applied to $f(\cdot, M)$ over $X_k^{(j)}$ satisfies $K(X_k^{(j)}; B_k) \subseteq \mathrm{int}(X_k^{(j)})$ with contraction factor $\rho_k^{(j)} < 1$. By Theorem 2, for every $\mu \in B_k$, there exists a unique zero $x^{(j)*}(\mu) \in X_k^{(j)}$.

(b) For each eigenvalue enclosure $\Lambda_k^{(j)}$: the Gershgorin disks of $Df_x(X_k^{(j)}, B_k)$ are contained in the stated enclosures, and the stability classification follows from the sign analysis of the enclosure real parts.

(c) The regime derivation D_k correctly applies an inference rule from Definition 4 to the stability types $\{s_k^{(1)}, \ldots, s_k^{(m_k)}\}$ to produce ℓ_k.

Therefore, for every $\mu \in B_k$, the regime claim ℓ_k is correct.

No contradictions. By hypothesis, $\mathrm{int}(B_i) \cap \mathrm{int}(B_j) = \emptyset$ for $i \neq j$. Therefore each $\mu \in \mathcal{P}_{\mathrm{cert}} = \bigcup_k B_k$ belongs to at most one cell interior, and any μ on a cell boundary belongs to cells that, by construction during refinement, are derived from the same parent and carry consistent labels.

Explicit gaps. The uncertified region $\mathcal{P} \setminus \mathcal{P}_{\mathrm{cert}}$ is explicitly reported. No regime label is assigned to these regions, so no incorrect claims are made. □

Remark 9. *The composition theorem does not require cells to cover* $\mathcal{P}$ *completely. The atlas soundness guarantee is: every certified cell is correct. The coverage fraction* $\mathrm{vol}(\mathcal{P}_{\mathrm{cert}})/\mathrm{vol}(\mathcal{P})$ *measures completeness but does not affect soundness.*

D GP Coverage Bounds via Borell-TIS Inequality

We derive probabilistic bounds on the GP-predicted uncovered volume using the Borell-TIS inequality [17].

D.1 Setup

Let $g : \mathcal{P} \to \mathbb{R}$ be a Gaussian process with Matérn-5/2 kernel trained on regime labels from certified cells. The GP posterior is $g(\mu) \mid \mathcal{D} \sim \mathcal{N}(m(\mu), \sigma^2(\mu))$ where $\mathcal{D} = \{(\mu_k, \ell_k)\}_{k=1}^K$ is the training data.

Define the *regime boundary indicator* $h(\mu) = |m(\mu) - \mathrm{round}(m(\mu))|$, measuring distance from the nearest integer (regime boundary in the GP encoding). Uncovered regions correspond to $h(\mu) < \tau$ for some threshold $\tau > 0$.

D.2 Borell-TIS Inequality

Theorem 10 (Borell-TIS [17])**.** *Let* g *be a centered Gaussian process on a compact set* T *with* $\sigma_T^2 = \sup_{t \in T} \mathbb{E}[g(t)^2] < \infty$*. Let* $u = \mathbb{E}[\sup_{t \in T} g(t)]$*. Then for all* $t > 0$*:*

$$\Pr\left[\sup_{t \in T} g(t) > u + t\right] \leq \exp\left(-\frac{t^2}{2\sigma_T^2}\right)$$

D.3 Application to Coverage Bounds

We apply Theorem 10 to bound the volume of parameter space where the GP predicts regime transitions that have not been certified.

Let $\tilde{g}(\mu) = g(\mu) - m(\mu)$ be the centered GP residual. The uncertified volume near predicted regime boundaries satisfies:

$$\text{vol}(\{\mu \in \mathcal{P} : \sigma(\mu) > \sigma_{\max}\}) \leq \text{vol}(\mathcal{P}) \cdot \Pr\left[\sup_{\mu} \tilde{g}(\mu) > u + (\sigma_{\max} - u)\right]$$

For the Matérn-5/2 kernel with length scales ℓ_d, the supremum mean can be bounded using Dudley's entropy integral [19]:

$$u = \mathbb{E}\left[\sup_{\mu \in \mathcal{P}} \tilde{g}(\mu)\right] \leq C \cdot \sigma_f \cdot \sqrt{\sum_{d=1}^{p} \log\left(\frac{\text{diam}(\mathcal{P}_d)}{\ell_d}\right)}$$

where C is a universal constant and σ_f is the GP signal variance.

D.4 Practical Stopping Criterion

After certifying K cells with coverage fraction $c = \text{vol}(\mathcal{P}_{\text{cert}})/\text{vol}(\mathcal{P})$, the Borell-TIS bound provides:

$$\Pr[\text{missed regime transition in uncertified region}] \leq \exp\left(-\frac{(\sigma_{\max} - u)^2}{2\sigma^2_{\mathcal{P}\setminus\mathcal{P}_{\text{cert}}}}\right)$$

where $\sigma^2_{\mathcal{P}\setminus\mathcal{P}_{\text{cert}}}$ is the maximum GP posterior variance in the uncertified region.

This bound decreases as:

- More cells are certified (reducing $\sigma^2_{\mathcal{P}\setminus\mathcal{P}_{\text{cert}}}$ via GP conditioning).
- The uncertified region shrinks (approaching bifurcation boundaries where regime transitions are expected).
- GP hyperparameters are well-calibrated (accurate σ_f and ℓ_d estimates).

The implementation evaluates this bound after each GP retraining and reports it as a confidence metric in the proof trace. We emphasize that this bound is *informational only*: it does not affect the soundness of certified cells, which rests solely on the Krawczyk operator and MiniCheck verification.

E Convergence Rate Analysis

Proposition 11 (Refinement Convergence Rate)**.** *Let $\Sigma \subset \mathcal{P}$ be the bifurcation manifold, assumed to be a piecewise-smooth $(p-1)$-dimensional submanifold with finite $(p-1)$-Hausdorff measure $\mathcal{H}^{p-1}(\Sigma)$. At refinement depth d, the uncertified volume fraction satisfies:*

$$\frac{\text{vol}(\textit{uncertified})}{\text{vol}(\mathcal{P})} = O\left(\frac{D \cdot \mathcal{H}^{p-1}(\Sigma)}{2^{d/p} \cdot \text{vol}(\mathcal{P})}\right)$$

where D is the domain diameter.

Proof. At depth d with isotropic splitting, each cell has diameter $D_d = D/2^{d/p}$ (splitting one dimension per level, cycling through p dimensions). The Krawczyk operator certifies any box B with $\text{dist}(B, \Sigma) > D_d \cdot \alpha$ for some constant α depending on the Lipschitz constants of f and Df. This is because, away from Σ, the eigenvalue gap $\gamma(\mu)$ is bounded below, and the Krawczyk contraction condition $\rho < 1$ is satisfied for sufficiently small boxes.

The uncertified volume is therefore contained in the $D_d \cdot \alpha$-neighborhood of Σ:

$$\text{vol}(\text{uncertified}) \leq \text{vol}(N_{D_d \alpha}(\Sigma)) = O(D_d \cdot \alpha \cdot \mathcal{H}^{p-1}(\Sigma))$$

by the tubular neighborhood volume formula for smooth submanifolds.

Substituting $D_d = D/2^{d/p}$:

$$\frac{\text{vol}(\text{uncertified})}{\text{vol}(\mathcal{P})} = O\left(\frac{D \cdot \alpha \cdot \mathcal{H}^{p-1}(\Sigma)}{2^{d/p} \cdot \text{vol}(\mathcal{P})}\right) \to 0 \quad \text{as } d \to \infty$$

□

Brusselator example. For the Brusselator with $p = 2$, the Hopf bifurcation curve $B = 1 + A^2$ has arc length ~ 2.5 in $[0.8, 2] \times [1.5, 3.5]$. At depth 5, $D_5 \approx 0.06$, giving expected uncertified fraction $\sim 0.06 \times 2.5/(1.2 \times 2.0) \approx 6\%$. The observed 14.1% gap (at depth 6 with 17 cells) is larger due to the Krawczyk operator requiring additional eigenvalue-gap margin near the bifurcation, inflating α.

Anisotropic refinement improvement. Anisotropic splitting guided by eigenvalue sensitivity (splitting along the dimension of maximum $w_d \cdot \sigma_d$ where σ_d is the per-dimension eigenvalue sensitivity) reduces the effective constant α by aligning cells with the bifurcation manifold. In the Brusselator, the Hopf boundary has slope $\partial B/\partial A = 2A$, so splitting along B at high A values concentrates cells where they are most needed.

F All 15 Model Definitions

We provide complete ODE definitions, parameter ranges, and bifurcation structure for all 15 benchmark models.

F.1 Toggle Switch [15]

A synthetic genetic circuit with two mutually repressing genes.

$n = 2$ state variables, $p = 4$ parameters (with Hill coefficients $n_1 = n_2 = 2$ fixed):

$$\dot{x}_1 = \frac{\alpha_1}{1 + x_2^{n_1}} - x_1$$
$$\dot{x}_2 = \frac{\alpha_2}{1 + x_1^{n_2}} - x_2$$

Parameters: $\alpha_1, \alpha_2 \in [2, 4]$. Bifurcation: saddle-node bifurcation separating monostable and bistable regimes. The model exhibits bistability when both α_i are sufficiently large.

F.2 Brusselator

A model for autocatalytic chemical reactions.

$n = 2$, $p = 2$:

$$\dot{x}_1 = A - (B+1)x_1 + x_1^2 x_2$$
$$\dot{x}_2 = Bx_1 - x_1^2 x_2$$

Parameters: $A \in [0.8, 2.0]$, $B \in [1.5, 3.5]$. Unique equilibrium at $(x_1^*, x_2^*) = (A, B/A)$. Hopf bifurcation at $B = 1 + A^2$: for $B < 1 + A^2$ the equilibrium is stable (monostable); for $B > 1 + A^2$ it is unstable and a limit cycle exists (oscillatory).

F.3 Sel'kov Glycolysis

A model for glycolytic oscillations via phosphofructokinase feedback.

$n = 2$, $p = 2$:

$$\dot{x}_1 = -x_1 + ax_2 + x_1^2 x_2$$
$$\dot{x}_2 = b - ax_2 - x_1^2 x_2$$

Parameters: $a \in [0.05, 0.15]$, $b \in [0.4, 0.8]$. The unique equilibrium is stable throughout the parameter box (monostable regime).

F.4 Repressilator [16]

A synthetic gene regulatory network of three transcriptional repressors forming a cyclic inhibition loop.

$n = 3$, $p = 4$ (with $K = 1$, Hill $n = 2$, $\gamma = 1$ fixed):

$$\dot{x}_1 = \frac{\alpha}{1 + x_3^2} - \gamma x_1$$
$$\dot{x}_2 = \frac{\alpha}{1 + x_1^2} - \gamma x_2$$
$$\dot{x}_3 = \frac{\alpha}{1 + x_2^2} - \gamma x_3$$

Parameter: $\alpha \in [3.5, 5.5]$. The symmetric equilibrium $x_1 = x_2 = x_3 = x^*$ loses stability via Hopf bifurcation as α increases, generating sustained oscillations.

F.5 Goodwin Oscillator [14]

A model for oscillatory enzymatic control with end-product inhibition.

$n = 3$, $p = 2$ (with $K = 1$, Hill coefficient $n = 4$ fixed):

$$\dot{x}_1 = \frac{a}{1 + x_3^4} - bx_1$$
$$\dot{x}_2 = x_1 - bx_2$$
$$\dot{x}_3 = x_2 - bx_3$$

Parameters: $a \in [3, 8]$, $b \in [0.8, 1.2]$. Oscillations emerge via Hopf bifurcation when the Hill coefficient is ≥ 8 (or effectively when a/b^3 exceeds a critical threshold with $n = 4$ fixed).

F.6 Lotka-Volterra Predator-Prey

The classical predator-prey model.

$n = 2$, $p = 3$ (with $\delta = 1$ fixed):

$$\dot{x} = \alpha x - \beta xy$$
$$\dot{y} = xy - \gamma y$$

Parameters: $\alpha \in [0.5, 1.5]$, $\beta \in [0.8, 1.2]$, $\gamma = 1$. Interior equilibrium at $(x^*, y^*) = (\gamma, \alpha/\beta)$ with center-type linearization (purely imaginary eigenvalues). The Krawczyk operator certifies existence but stability classification is inconclusive due to the center.

F.7 FitzHugh-Nagumo Neuron Model

A simplified model of neuronal excitability.

$n = 2$, $p = 2$ (with $a = 0.7$, $b = 0.8$ fixed):

$$\dot{v} = v - \frac{v^3}{3} - w + I$$
$$\dot{w} = \epsilon(v + 0.7 - 0.8w)$$

Parameters: $I \in [0, 1]$ (external current), $\epsilon \in [0.05, 0.2]$ (time-scale separation). Exhibits excitable-to-oscillatory transition via supercritical Hopf bifurcation as I increases past the critical current I_c.

F.8 Tyson Cell Cycle [20]

A minimal model of the cdc2-cyclin interaction governing the eukaryotic cell division cycle.

$n = 2$, $p = 2$:

$$\dot{u} = k_1 - (k_2 + k_2' u)v$$
$$\dot{v} = k_4(u - v)(1 + Kv^2) - k_6 v$$

Parameters: $k_1 \in [0.01, 0.05]$, $k_6 \in [0.5, 2.0]$, with $k_2 = 0$, $k_2' = 1$, $k_4 = 0.5$, $K = 10$ fixed. The model exhibits bistability and hysteresis, with the cell cycle "switch" controlled by the balance of synthesis (k_1) and degradation (k_6).

F.9 MAPK Cascade [21]

A reduced model of the mitogen-activated protein kinase signaling cascade exhibiting ultrasensitive switching.

$n = 2$, $p = 2$:

$$\dot{x}_1 = \frac{V_1(1 - x_1)}{K_1 + (1 - x_1)} - \frac{V_2 x_1}{K_2 + x_1}$$
$$\dot{x}_2 = \frac{V_3 x_1(1 - x_2)}{K_3 + (1 - x_2)} - \frac{V_4 x_2}{K_4 + x_2}$$

Parameters: $V_1 \in [0.5, 2.0]$, $K_1 \in [0.01, 0.1]$, with $V_2 = 1$, $K_2 = 0.05$, $V_3 = 1$, $K_3 = 0.05$, $V_4 = 1$, $K_4 = 0.05$ fixed. Monostable throughout the parameter range, with the unique equilibrium tracking the ultrasensitive input-output curve.

F.10 p53-Mdm2 Feedback Loop [22]

A model of the p53 tumor suppressor and its negative regulator Mdm2, exhibiting damped oscillations in response to DNA damage.

$n = 2$, $p = 2$:

$$\dot{p} = s_p - d_p \cdot p - \frac{\beta \cdot p \cdot m}{K_m + p}$$
$$\dot{m} = s_m + \frac{\alpha \cdot p^2}{K_p^2 + p^2} - d_m \cdot m$$

Parameters: $s_p \in [0.5, 2.0]$ (p53 synthesis rate), $d_m \in [0.1, 0.5]$ (Mdm2 degradation rate), with $d_p = 0.2$, $\beta = 1$, $K_m = 0.5$, $s_m = 0.1$, $\alpha = 2$, $K_p = 1$ fixed. The rational RHS arises from Michaelis-Menten degradation kinetics.

F.11 Calcium Oscillator

A model of intracellular calcium oscillations mediated by IP3-sensitive stores.

$n = 2$, $p = 2$:

$$\dot{c} = v_0 + v_1\beta - kc + \frac{V_M c^2}{K^2 + c^2}$$
$$\dot{s} = v_s \frac{c^2}{K_s^2 + c^2} - v_r \frac{s}{K_r + s}$$

Parameters: $\beta \in [0.1, 0.8]$ (IP3 stimulation), $v_0 \in [0.5, 2.0]$ (basal influx), with $v_1 = 1$, $k = 0.5$, $V_M = 2$, $K = 0.5$, $v_s = 1$, $K_s = 0.5$, $v_r = 1$, $K_r = 0.5$ fixed. Exhibits a supercritical Hopf bifurcation from monostable to oscillatory as β increases.

F.12 Circadian Clock [23]

A simplified model of circadian oscillations in the *Drosophila* PER protein.

$n = 2$, $p = 2$:

$$\dot{M} = v_s \frac{K_I^n}{K_I^n + P^n} - v_m \frac{M}{K_m + M}$$
$$\dot{P} = k_s M - v_d \frac{P}{K_d + P}$$

Parameters: $v_s \in [0.5, 2.0]$ (maximal transcription rate), $v_d \in [0.5, 2.0]$ (maximal degradation rate), with $K_I = 1$, $n = 4$, $v_m = 0.65$, $K_m = 0.5$, $k_s = 0.38$, $K_d = 0.2$ fixed. Sustained oscillations arise when v_s is sufficiently large relative to v_d.

F.13 Genetic Oscillator

A three-gene oscillator with cascaded repression.

$n = 3$, $p = 2$:

$$\dot{x}_1 = \frac{a}{1 + x_3^h} - bx_1$$
$$\dot{x}_2 = \frac{a}{1 + x_1^h} - bx_2$$
$$\dot{x}_3 = x_2 - bx_3$$

Parameters: $a \in [2, 6]$, $b \in [0.5, 1.5]$, with Hill coefficient $h = 3$ fixed. Similar to the repressilator but with a direct coupling from x_2 to x_3 instead of Hill-function repression, altering the bifurcation structure.

F.14 Mutual Activation

A gene regulatory network with two mutually activating genes, exhibiting bistability via positive feedback.

$n = 2$, $p = 2$:

$$\dot{x}_1 = \frac{\alpha x_2^n}{K^n + x_2^n} - \gamma x_1$$
$$\dot{x}_2 = \frac{\alpha x_1^n}{K^n + x_1^n} - \gamma x_2$$

Parameters: $\alpha \in [2, 6]$, $\gamma \in [0.5, 1.5]$, with $K = 1$, $n = 2$ fixed. Bistability arises when α/γ exceeds a critical threshold. The model has both a symmetric equilibrium ($x_1 = x_2$) and, in the bistable regime, two asymmetric equilibria.

F.15 van der Pol Oscillator [24]

The classical relaxation oscillator.

$n = 2$, $p = 2$:

$$\dot{x} = y$$
$$\dot{y} = \mu(1 - x^2)y - x$$

Parameters: $\mu \in [0.5, 3.0]$, with the second parameter being a trivial scaling (fixed at 1). The unique equilibrium at the origin is always unstable for $\mu > 0$, with a stable limit cycle. The Krawczyk operator certifies the equilibrium at $(0, 0)$ with $\bar{\rho} = 0.000$ (trivial zero of a polynomial system), and the eigenvalue enclosure confirms instability.

G Comprehensive Ablation Tables

G.1 Tier-by-Tier Ablation

The tier-by-tier ablation (Table 5) confirms that Z3 verification (Tiers 2 and 3) does not affect coverage or certification decisions. Its role is to provide independent formal verification: a second opinion from an SMT solver that the interval-arithmetic-based certification is correct. Disabling Z3 saves 2–8% wall time (the Z3 overhead per cell is 13–65 ms).

G.2 GP-Guided vs. Uniform Refinement

At low parameter dimensions ($p \leq 4$), GP-guided refinement provides comparable performance to uniform refinement (Table 6). The GP's ECE is 0.0 across all models, indicating well-calibrated predictions at available sample sizes. High LOO error for single-cell models (Sel'kov, van der Pol) reflects the inherent difficulty of regression with one training point.

The GP's benefit is in prioritizing boundary cells: for the mutual activation model, GP-guided refinement achieves 78% vs. 75% coverage, as the GP identifies the saddle-node boundary and focuses cells there.

Table 5: Ablation: effect of disabling verification tiers. "Full" = all three tiers. "No Z3" = Tier 1 only (MiniCheck). Coverage and soundness are unaffected; only the verification confidence level changes.

Model	Full (Tier 1+2+3)		Tier 1 Only		Δ Coverage
	Coverage	Time	Coverage	Time	
Toggle switch	100%	3.6 s	100%	3.4 s	0%
Brusselator	85.9%	66.3 s	85.9%	63.5 s	0%
Sel'kov	100%	1.0 s	100%	0.9 s	0%
Repressilator	100%	0.7 s	100%	0.6 s	0%
Goodwin	100%	0.1 s	100%	0.1 s	0%
Lotka-Volterra	100%	0.2 s	100%	0.2 s	0%
FitzHugh-Nagumo	81.2%	19.3 s	81.2%	17.8 s	0%
Tyson cell cycle	100%	0.2 s	100%	0.2 s	0%
MAPK cascade	100%	<0.1 s	100%	<0.1 s	0%
p53-Mdm2	100%	1.5 s	100%	1.3 s	0%
Calcium oscillator	100%	4.6 s	100%	4.2 s	0%
Circadian clock	100%	3.3 s	100%	3.0 s	0%
Genetic oscillator	100%	0.1 s	100%	0.1 s	0%
Mutual activation	90.6%	51.2 s	90.6%	48.5 s	0%
van der Pol	100%	<0.1 s	100%	<0.1 s	0%

G.3 Newton-Guided vs. Bisection Search

Newton-guided search is essential for $n = 3$: bisection requires $O(2^{3d})$ subdivisions per depth level, making it impractical at depths beyond 3 (Table 7). For $n = 2$, Newton-guided search provides a modest speedup (4–7%) by reducing the number of bisection steps needed to locate equilibria.

H MiniCheck Design

MiniCheck is a standalone certificate verifier (1,000 LoC) depending only on Python and NumPy. It is the sole trusted computing base (TCB): all soundness guarantees ultimately rest on MiniCheck's independent verification.

H.1 Architecture

MiniCheck is structured in five modules:

1. *Interval arithmetic* (∼300 LoC): Classes `IV` (scalar interval), `IVec` (interval vector), `IMat` (interval matrix) with outward rounding and conservative ϵ-inflation of 10^{-15} relative to the magnitude at each operation.

2. *Model RHS* (∼310 LoC): Hand-coded RHS $f(x, \mu)$ and Jacobian $Df_x(x, \mu)$ for all 15 benchmark models. Each model is implemented as a standalone function with no symbolic computation or automatic differentiation—the Jacobian is computed by direct differentiation of the RHS formula.

3. *Krawczyk recomputation* (∼200 LoC): Independent Krawczyk operator implementation, containment check $K(X; M) \subseteq \text{int}(X)$, and contraction factor computation. Gershgorin disk eigenvalue enclosure for stability classification.

Table 6: GP-guided vs. uniform refinement (depth 4, 200 cells max).

Model	Uniform		GP-Guided		GP ECE	GP LOO
	Coverage	Time	Coverage	Time		
Toggle switch	100%	3.2 s	100%	3.1 s	0.000	0.002
Brusselator	50%	36.1 s	50%	37.8 s	0.000	9.336
Sel'kov	100%	0.8 s	100%	0.8 s	0.000	25.0
FitzHugh-Nagumo	62.5%	6.8 s	50%	8.2 s	0.000	25.0
Tyson cell cycle	100%	0.2 s	100%	0.2 s	0.000	0.001
MAPK cascade	100%	<0.1 s	100%	<0.1 s	0.000	0.001
Calcium oscillator	100%	3.8 s	100%	3.5 s	0.000	0.010
Mutual activation	75%	32.1 s	78%	34.5 s	0.000	5.210
van der Pol	100%	<0.1 s	100%	<0.1 s	0.000	0.000

Table 7: Newton vs. bisection on all 3-state models and selected 2-state models.

Model	n	Bisection		Newton-Guided	
		Coverage	Time	Coverage	Time
Repressilator	3	25.0%	69.8 s	100%	0.8 s
Goodwin	3	0.0%	166.6 s	100%	0.1 s
Genetic oscillator	3	0.0%	143.2 s	100%	0.1 s
Brusselator	2	85.9%	70.1 s	85.9%	66.3 s
FitzHugh-Nagumo	2	81.2%	20.1 s	81.2%	19.3 s

4. *Verification pipeline* (∼140 LoC): Certificate parsing (JSON format), equilibrium verification, stability verification, and regime label consistency checking. Reports PASS/FAIL with detailed diagnostics for each certificate component.

5. *CLI* (∼50 LoC): Command-line interface accepting a certificate file and returning exit code 0 (all checks pass) or 1 (any check fails).

H.2 Independence Guarantees

MiniCheck shares no code with the main PhaseCartographer pipeline:

- The interval arithmetic implementation is written from scratch, not imported from the main `interval/` module.
- The model RHS and Jacobian functions are hand-coded, not generated by the symbolic engine.
- The Krawczyk operator is reimplemented independently.
- The eigenvalue enclosure uses only Gershgorin disks (not the Bauer-Fike method used in the main pipeline).

This independence means that a bug in the main pipeline's interval arithmetic, symbolic differentiation, or Krawczyk implementation cannot produce a certificate that passes MiniCheck, unless the same bug exists independently in MiniCheck's separate implementation.

H.3 Verification Protocol

For each certificate entry, MiniCheck performs:

1. Parse the certificate JSON, extracting parameter box M, equilibrium enclosures $\{X^{(j)}\}$, contraction factors $\{\rho^{(j)}\}$, eigenvalue enclosures $\{\Lambda^{(j)}\}$, and regime label ℓ.
2. For each equilibrium enclosure $X^{(j)}$:
   (a) Recompute the Krawczyk operator $K(X^{(j)}; M)$ using MiniCheck's own interval arithmetic and model RHS/Jacobian.
   (b) Verify $K(X^{(j)}; M) \subseteq \text{int}(X^{(j)})$.
   (c) Verify $\rho^{(j)} < 1$ and check that MiniCheck's independent ρ agrees within tolerance.
3. For each eigenvalue enclosure $\Lambda^{(j)}$:
   (a) Compute Gershgorin disks of $Df_x(X^{(j)}, M)$.
   (b) Verify that the stated eigenvalue enclosures are contained in the Gershgorin disks.
   (c) Classify stability from the verified enclosures.
4. Re-derive the regime label from the verified stability types using the inference rules (Definition 4). Verify consistency with the stated label ℓ.

I Proof Trace Format Specification

PhaseCartographer exports proof traces in a structured JSON format with hash-chain integrity verification, enabling independent auditing and replay.

I.1 Trace Structure

A proof trace $\mathcal{T}$ is a sequence of entries $\mathcal{T} = (e_1, e_2, \ldots, e_N)$ where each entry e_i has the following fields:

```
{
  "index": <integer>,
  "timestamp": <ISO 8601>,
  "type": "certify" | "split" | "z3_verify" | "gp_train",
  "cell_id": <string>,
  "parent_id": <string or null>,
  "parameter_box": [[lo_1, hi_1], ..., [lo_p, hi_p]],
  "equilibria": [
    {
      "state_box": [[lo_1, hi_1], ..., [lo_n, hi_n]],
      "contraction_factor": <float>,
      "eigenvalue_enclosures": [
        {"real": [lo, hi], "imag": [lo, hi]}, ...
      ],
      "stability": "stable" | "unstable" | "saddle"
                   | "center" | "inconclusive"
```

```
    }, ...
  ],
  "regime": "monostable" | "bistable" | ...,
  "regime_derivation": {
    "rule": "MONO" | "BI" | "EXC" | ...,
    "premises": [...],
    "conclusion": <string>
  },
  "tier_results": {
    "tier1": "PASS" | "FAIL",
    "tier2": "CONFIRMED" | "INCONCLUSIVE" | "N/A",
    "tier3": "CONFIRMED" | "INCONCLUSIVE" | "N/A"
  },
  "hash": <SHA-256 hex string>,
  "prev_hash": <SHA-256 hex string or null>
}
```

I.2 Hash-Chain Integrity

Each entry's hash is computed as:

$$h_i = \text{SHA256}(h_{i-1} \,\|\, \text{canonical}(e_i \setminus \{\text{hash}, \text{prev_hash}\}))$$

with $h_0 = \text{SHA256}(\text{“PhaseCartographer-v1”})$. The canonical form sorts JSON keys alphabetically and normalizes floating-point numbers to 15 significant digits.

This hash chain ensures:

- *Tamper detection:* Any modification to a past entry invalidates all subsequent hashes.
- *Ordering:* The chain enforces a total order on certification events.
- *Completeness:* The final hash commits to the entire certification history.

I.3 Replay Verification

An independent verifier can replay the proof trace by:

1. Parsing each entry e_i in sequence.
2. Recomputing the Krawczyk operator and eigenvalue enclosures (as MiniCheck does).
3. Verifying the hash chain: $h_i = \text{SHA256}(h_{i-1} \,\|\, \text{canonical}(e_i))$.
4. Checking that the final atlas (union of all “certify” entries) satisfies the composition theorem (Theorem 5).

J Extended Z3 Verification Analysis

We provide detailed Z3 verification statistics for all 15 models.

J.1 Equilibrium Existence Verification

For each certified cell, we encode the formula $\exists x \in X, \mu \in M : f(x, \mu) = 0$ in the QF_NRA theory and check satisfiability. For polynomial models, this is exact. For Hill-function and rational models, we clear denominators to obtain a polynomial system, which is equisatisfiable over the domain where denominators are nonzero (verified by the interval enclosure).

Table 8: Z3 equilibrium existence verification: per-model statistics.

Model	Cells	Z3 Result	Median Time	Notes
Toggle switch	2	2/2 SAT	22 ms	Hill cleared to poly
Brusselator	17	17/17 SAT	18 ms	Direct polynomial
Sel'kov	1	1/1 SAT	32 ms	Direct polynomial
Repressilator	1	1/1 SAT	45 ms	Hill cleared to poly
Goodwin	2	2/2 SAT	38 ms	Hill cleared to poly
Lotka-Volterra	1	1/1 SAT	10 ms	Direct polynomial
FitzHugh-Nagumo	15	15/15 SAT	15 ms	Direct polynomial
Tyson cell cycle	1	1/1 SAT	28 ms	Direct polynomial
MAPK cascade	1	1/1 SAT	41 ms	Rational cleared
p53-Mdm2	2	2/2 SAT	35 ms	Rational cleared
Calcium oscillator	3	3/3 SAT	24 ms	Rational cleared
Circadian clock	4	4/4 SAT	29 ms	Hill cleared to poly
Genetic oscillator	2	2/2 SAT	33 ms	Hill cleared to poly
Mutual activation	7	7/7 SAT	26 ms	Hill cleared to poly
van der Pol	1	1/1 SAT	5 ms	Trivial $(0, 0)$
Total	**60**	**60/60**	24 ms	

All 60 cells produce SAT results with algebraic witnesses (Table 8). Median verification time is 24 ms per cell. The van der Pol oscillator is fastest (5 ms) because the equilibrium at the origin makes the polynomial system trivially satisfiable.

J.2 Stability Verification via Routh-Hurwitz

For stability, we encode the Routh-Hurwitz conditions in QF_NRA. For $n = 2$: the characteristic polynomial of $J = Df_x(x^*, \mu)$ is $\lambda^2 - \mathrm{tr}(J)\lambda + \det(J)$; stability requires $\mathrm{tr}(J) < 0$ and $\det(J) > 0$. We ask Z3 to verify UNSAT for:

$$\exists (x, \mu) \in X \times M : \|f(x, \mu)\|_\infty < \delta \ \wedge \ (\mathrm{tr}(Df_x) \geq 0 \ \vee \ \det(Df_x) \leq 0)$$

Z3 confirms stability (or instability) in 34 of 60 cells (Table 9). The 11 inconclusive results fall into two categories:

1. *Near-bifurcation cells* (10 cells): The Routh-Hurwitz condition $\mathrm{tr}(J) < 0$ or $\det(J) > 0$ is *not* universally satisfied within the cell because the cell straddles a Hopf or saddle-node boundary. Z3 correctly reports SAT (a counterexample exists within the cell).

2. *Center-type equilibria* (1 cell, Lotka-Volterra): $\mathrm{tr}(J) = 0$ exactly, so stability cannot be confirmed or denied via Routh-Hurwitz.

The remaining 15 cells are N/A because their Hill-function or rational Jacobians produce non-polynomial Routh-Hurwitz conditions that cannot be directly encoded in QF_NRA without denominator clearing, which we do not attempt for stability verification to avoid introducing spurious solutions.

Table 9: Z3 stability verification: per-model results.

Model	Cells	Confirmed	Inconclusive	Reason for Inconclusive
Toggle switch	2	—	—	Hill: non-polynomial Jacobian
Brusselator	17	8	9	Hopf boundary $B = 1 + A^2$
Sel'kov	1	1	0	Well-separated from bifurcation
Repressilator	1	—	—	Hill: 3-state RH not polynomial
Goodwin	2	—	—	Hill: 3-state RH not polynomial
Lotka-Volterra	1	—	—	Center: $\mathrm{tr}(J) = 0$ exactly
FitzHugh-Nagumo	15	14	1	One cell at Hopf boundary
Tyson cell cycle	1	1	0	Stable throughout box
MAPK cascade	1	—	—	Rational Jacobian
p53-Mdm2	2	—	—	Rational Jacobian
Calcium oscillator	3	3	0	Polynomial, well-separated
Circadian clock	4	—	—	Hill: non-polynomial Jacobian
Genetic oscillator	2	—	—	Hill: 3-state RH
Mutual activation	7	6	1	One cell at saddle-node boundary
van der Pol	1	1	0	Unstable confirmed ($\mathrm{tr} > 0$)
Total	**60**	**34**	**11**	15 cells N/A (non-poly)

J.3 Completeness Checking

For polynomial models where Z3 equilibrium verification succeeds, we additionally check whether all equilibria have been found by asking:

$$\exists x \in D \setminus \bigcup_j X^{(j)}, \mu \in M : f(x, \mu) = 0$$

where D is the state-space search domain and $X^{(j)}$ are the certified equilibrium enclosures. An UNSAT result confirms that no equilibria exist outside the known enclosures.

This completeness check succeeds for 4 of 8 polynomial models (Sel'kov, Tyson, van der Pol, calcium oscillator). For the remaining 4 (Brusselator, FitzHugh-Nagumo, Lotka-Volterra, MAPK), Z3 times out on the completeness query, which involves quantifying over the complement of a union of boxes—a harder problem than existence in a single box.

K Depth Scaling Results

We evaluate the Brusselator model at increasing refinement depths with `max_cells=1000` to validate the convergence rate analysis (Proposition 11).

At depth 2, the parameter cells are too coarse to certify any box: every cell straddles the Hopf boundary $B = 1 + A^2$, causing the Krawczyk operator to fail. At depth 3, three cells are certified in the interior of the stable regime (below the Hopf curve), achieving 37.5% coverage. Coverage increases monotonically through depth 5 (84.4%) and depth 6 (85.9%), with diminishing returns as the remaining uncertified cells are concentrated at the bifurcation boundary. The transition from depth 4 to depth 5 produces the largest coverage gain (34.4 percentage points), consistent with cells at that scale becoming small enough to avoid straddling the Hopf curve.

Table 10: Brusselator depth scaling (`max_cells=1000`). Coverage increases monotonically with depth, consistent with $O(2^{-d/p})$ convergence.

Depth	Cells	Coverage	Time
2	0	0.0%	10.5 s
3	3	37.5%	20.4 s
4	5	50.0%	33.3 s
5	16	84.4%	52.7 s
6	17	85.9%	66.3 s

L Empirical δ-Soundness Validation

We empirically validate the δ-soundness bound (Theorem 3) by evaluating the condition $\delta_{\text{solver}} < \gamma/(\|R\| \cdot L_{Df} \cdot \text{rad}(X \times M) + 1)$ on cells where Z3 stability verification succeeds.

Table 11: Empirical δ-soundness validation. "Gap > 0" indicates cells where the δ-soundness condition is strictly satisfied. All such cells are sound (correct stability classification).

Model	Tested	Gap > 0	All Sound	Min Gap
Brusselator	5	3	Yes	—
Calcium oscillator	3	3	Yes	0.6402
FitzHugh-Nagumo	3	1	Yes	—

For the Brusselator, 3 of 5 tested cells have a positive δ-soundness gap; the remaining 2 cells are near the Hopf boundary where $\gamma \to 0$ makes the bound tight. The calcium oscillator achieves the largest margins, with all 3 cells satisfying the condition and a minimum gap of 0.6402, reflecting the well-separated eigenvalues in this model. For FitzHugh-Nagumo, 1 of 3 tested cells satisfies the bound; the other 2 are near the excitable-oscillatory transition where eigenvalue enclosures approach zero. In all cases, no cell with a positive gap produces an incorrect stability classification, confirming that the bound from Theorem 3 is both sufficient and practically useful.

M User-Defined Model Input

While the evaluation in Section 9 uses 18 built-in benchmark models, the certification pipeline is fully generic: any ODE system $\dot{x} = f(x, \mu)$ expressible as a symbolic right-hand side can be analyzed. To make this accessible without Python programming, we provide a YAML-based model file format and corresponding CLI commands.

M.1 Model File Format

A user-defined model specifies state variable names, parameter names with search bounds, SymPy-parseable RHS expressions (one per state), state-space bounds for equilibrium search, and optional metadata (RHS type, expected regimes, description). The parser constructs a `SymbolicRHS` instance and wraps it in a `BenchmarkModel` dataclass, making user models indistinguishable from built-in benchmarks in the certification pipeline.

M.2 CLI Workflow

Three commands support the user-defined model workflow:

1. `init-model`: generates a starter YAML template (2D, 3D, or oscillator).
2. `check`: parses and validates a model file without running analysis, reporting any issues (missing fields, undeclared symbols, dimension mismatches).
3. `--file`: runs the full certification pipeline on a user-defined model, producing the same certified phase atlas, Z3 verification, and δ-soundness analysis as built-in benchmarks.

The `check` command enables a fast edit-validate loop before committing to the full (potentially expensive) certification run. All three CLI commands, the YAML format, and the programmatic `load_model_file()` and `run_model()` APIs are documented in `API.md`.